MCQ 1: understand and choose

Five single-answer questions for practice. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official EAM format.

1. 180° equals
   - A. 2π
   - B. π
   - C. $\frac{\pi}{2}$
   - D. 1
2. $\cos(\frac{\pi}{3}) =$
   - A. $\frac{1}{2}$
   - B. $\frac{\sqrt{3}}{2}$
   - C. 0
   - D. 1
3. $\sin(-\frac{\pi}{2}) =$
   - A. 1
   - B. 0
   - C. $\frac{1}{2}$
   - D. -1
4. $\cos^2 x + \sin^2 x =$
   - A. 0
   - B. 2
   - C. 1
   - D. x
5. The same point corresponds to angles differing by
   - A. π
   - B. $2k\pi$, k an integer
   - C. $\frac{\pi}{2}$
   - D. 1

MCQ 2: understand and choose

Five single-answer questions for practice. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official EAM format.

1. $\cos(\pi - x) =$
   - A. $\cos x$
   - B. $\sin x$
   - C. $-\cos x$
   - D. $-\sin x$
2. $\sin(\frac{\pi}{4}) =$
   - A. $\frac{\sqrt{2}}{2}$
   - B. $\frac{1}{2}$
   - C. $\frac{\sqrt{3}}{2}$
   - D. $\sqrt{2}$
3. Image of $\frac{3\pi}{2}$:
   - A. $(1, 0)$
   - B. $(-1, 0)$
   - C. $(0, 1)$
   - D. $(0, -1)$
4. Arc of radius 3 and angle 2 radians: length
   - A. $\frac{2}{3}$
   - B. 6
   - C. $\frac{3}{2}$
   - D. 6π
5. Interior of the second quadrant: sign of the cosine
   - A. positive
   - B. zero everywhere
   - C. negative
   - D. cannot be determined

True or false? Justify.

1. $\cos(-x) = -\cos x$ for all x.
2. $\sin(x + 2\pi) = \sin x$.
3. The value of $\sin x$ always determines $\cos x$.

Key point · Self-assessment. I reduce angles by complete turns; I read cosine and sine in the right order; I justify the signs; I keep the exact values.

For each objective, indicate: achieved without help; achieved with help; to be reworked. Write one example and one mistake to avoid.